

cc:
X U.S. EPA-Region V
Cadillac District
X SWQAS (toxics review, Walker)
SWQAS (conventionals review)
X P-File

Name: Diane Carlson
 Surface Water Permits Section
 Water Division
 Date: 4/14/2003
 Revised: 6/14/04

BASIS FOR DECISION MEMO

Applicant: City of Cadillac
 Facility: Cadillac WWTP
 1121 Plett Road
 Cadillac, Michigan 49601

Permit No. MI0020257
 PID 21871.1

Monitoring Point 001A: Authorization to discharge treated municipal wastewater through Monitoring Point 001A to Outfall 001. Outfall 001 discharges to the Clam River. Rated design capacity is 3.2 MGD.

<u>Parameters</u>	<u>Maximum Monthly</u>	<u>Maximum 7-Day</u>	<u>Maximum Daily</u>	<u>Basis for Limits</u>	
Carbonaceous Biochemical Oxygen Demand (CBOD5)					
May 1 – Nov 30	7	---	10	WQBEL	
Dec 1 – Mar 31	25	---	34	WQBEL	
Apr 1 – Apr 30	20	---	30	WQBEL	
Total Suspended Solids					
May 1 – Nov 30	20	30	---	PWJ	
Dec 1 – – Apr 30	30	45	---	STS	
Ammonia Nitrogen (as N)					
May 1 – Sept 30	0.9	---	2 mg/l	WQBEL	
Oct 1 – Nov 30	1.9	---	2 mg/l	WQBEL	
Dec 1 – Mar 31	4.2	---	---	WQBEL	
Apr 1 – Apr 30	5.5	---	---	WQBEL	
Total Phosphorus (as P)					
May 1 – Mar 31	0.5	---	---	WQBEL	
Apr 1 – – Apr 30	0.65	---	---	WQBEL	
Fecal Coliform Bacteria	200	400	---	WQBEL	
Total Mercury (Begin 7/1/2005)	30	---	---	WQBEL	
			Minimum Monthly		
Total Suspended Solids Minimum % Removal	---		85%	STS	
			Minimum Daily	Maximum Daily	
pH	---	6.5	---	9.0	WQBEL
Dissolved Oxygen					
May 1 – Nov 30		6.0	---	---	WQBEL
Dec 1 – Apr 30		5.0	---	---	WQBEL

Facility flow is 3.2 mgd or 4.65 cfs. 95% exceedance flow is 2.1 cfs.
Dilution ratio is $3.2/(4.65+3.2) = 0.40$. Require acute toxicity and chronic toxicity tests.

Storm water is collected, but not treated. A program is required.

A mercury minimization program is required to meet the mercury limit. Since this community has a pretreatment program, much of the reduction may be accomplished within the IPP. Mercury: Monthly monitoring and a limit go into effect in one year. Until that time, quarterly monitoring is required. Monitoring frequency reduction is proposed to allow frequency reduction should our monthly monitoring policy change for this low level discharge.

Copper, cadmium, and zinc quarterly monitoring are required.
The plant will be converting to uv disinfection in the near future. TRC monitoring may be eliminated when chlorine is no longer used.

Lindane. The Lindane limit is based on eight values and 86 nondetects reported in DMRs. Monthly monitoring is required until January 1, 2006. A limit and twice monthly monitoring is in effect thereafter.

Chronic toxicity: The theoretical chronic limit is 1.1 TU_C. From 1998 to January of 2003, three tests showed measurable toxicity. One test was determined to be a valid test for limit development. Using the GLI WET rules promulgated by EPA, a limit was proposed for this discharge.

Biosolids are land applied.

The permit was contested November 24, 2003. Additional operating status, mercury, chronic toxicity, and Lindane data was submitted April 19, 2004. Based on recent elimination of chlorine disinfection equipment, more recent toxicity tests were used to assess the discharge. A TRC limit and chronic toxicity limit are no longer required; however, quarterly toxicity monitoring will continue to verify improved operations. Additional Lindane data supported elimination of limits and addition of quarterly monitoring.

Mercury limits are still required; however, the effective date for limits is based on the resolution date for the contested issues. Storm water dates were corrected to the new effective date also: 7/1/2004.

Key

WQBEL - Water Quality Based Effluent Limit
PWJ - Permit Writer's Judgment
STS - Secondary Treatment Standards
WQS - Water Quality Standards